

6.6 LINEAR DEPENDENCE OF VECTORS

A set of vectors $\mathbf{x}^1, \mathbf{x}^2, \dots, \mathbf{x}^n$ are said to be **linearly independent** if there exists scalars $c_1, c_2, \dots, c_n$ not all zero such that $c_1\mathbf{x}^1 + c_2\mathbf{x}^2 + \dots + c_n\mathbf{x}^n = \mathbf{0}$.

On the contrary, if no such scalars exist, then the vectors $\mathbf{x}^1, \dots, \mathbf{x}^n$ are linearly independent.

The term ‘linearly’ in the above definition is important because only linear operations, i.e., scalar multiplication and vector addition, are permitted in obtaining the null vector.

Two important results regarding linear independence or lack of it are as follows:

- If $\mathbf{x}^1, \mathbf{x}^2, \dots, \mathbf{x}^k$ a subset of a set of vectors $\mathbf{x}^1, \mathbf{x}^2, \dots, \mathbf{x}^n$ ($n > k$), are linearly dependent, then the entire set is linearly dependent.
- If a set of vectors $\mathbf{x}^1, \mathbf{x}^2, \dots, \mathbf{x}^n$ are linearly independent, then any subset of vectors, say, $\mathbf{x}^1, \mathbf{x}^2, \dots, \mathbf{x}^k$ ($k < n$) are also linearly independent.

Example: For the two vectors $x = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}$ and $y = \begin{pmatrix} 3 \\ 0 \\ 9 \end{pmatrix}$

we have, $3x - y = \mathbf{0}$. Therefore, the two vectors x and y are linearly dependent.

Also, from the equation, we get $y = 3x$

In other words, the vector y can be obtained as scalar multiple of vector x .

In general, if k vectors $\mathbf{x}^1, \mathbf{x}^2, \dots, \mathbf{x}^k$ are linearly dependent, then any one of them, having non-zero coefficient in the dependence equation, can be obtained as a linear combination of the rest.

Let us consider another example:

$$x = \begin{pmatrix} 1 \\ 3 \end{pmatrix}, y = \begin{pmatrix} -2 \\ 5 \end{pmatrix}, \text{ and } z = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$$

$$\text{so, } x - y - z = \mathbf{0}$$

Equivalently,

$$z = x - y$$

In other words, the vector z can be obtained as scalar multiple of the other two vectors x and y .

In each of these cases any of the given vectors can be obtained from the remaining vectors through scalar multiplication and vector addition (or subtraction); hence, vectors are linearly dependent.

6.7 GENERATORS AND BASIS

Consider the case of two-dimensional Euclidian space (E^2).

[Note: An n -dimensional Euclidian space (Euclidian vector space) is defined as the collection of all vectors (points) $a = (a_1, a_2, \dots, a_n)$, for $a_i \in \mathbb{R}$, along with the operations of vector addition and scalar multiplication; and with the concept of distance between the vectors]

In E^2 , for the vectors $\mathbf{u} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, it is easily seen that any vector $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ may be generated from the vectors $\mathbf{u}$ and $\mathbf{v}$ as follows

$$\mathbf{x} = x_1\mathbf{u} + x_2\mathbf{v}, \text{ for } x_1, x_2 \in \mathbb{R}.$$

Such vectors $\mathbf{u}$ and $\mathbf{v}$ —from which all other vectors can be obtained as linear combinations as above—are called ‘**generators**’ of the vector space, in this case, of the two-dimensional spaces (E^2).

Example: Consider the vector $\begin{pmatrix} 2 \\ 5 \end{pmatrix}$. This can be expressed as

$$\begin{aligned} 2\begin{pmatrix} 1 \\ 0 \end{pmatrix} + 5\begin{pmatrix} 0 \\ 1 \end{pmatrix}, & \quad \text{or } 2\begin{pmatrix} 1 \\ 1 \end{pmatrix} + 3\begin{pmatrix} 0 \\ 1 \end{pmatrix}, \\ & \quad \text{or } \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} + 4\begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ & \quad \text{or even } 2\begin{pmatrix} 1 \\ 2 \end{pmatrix} + 1\begin{pmatrix} 0 \\ 1 \end{pmatrix} + 3\begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{aligned}$$

Thus, we can generate the vector $\begin{pmatrix} 2 \\ 5 \end{pmatrix}$ in a variety of ways by taking a set of vectors such as $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$, **or** $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$, **or** $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$, **or** $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$.

Each of these pairs of these combinations of vectors like $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$, $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ is called a generator of E^2 .

Let us discuss some important points that have to be kept in mind.

- First, an arbitrary choice of $\mathbf{u}$, $\mathbf{v}$ etc. may not be good enough to generate all vectors $\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$
- Second, in the case of E^2 , we need at least two vectors for this purpose. No single vector is, by itself, capable of giving us all elements of E^2 . In general, for the Euclidean space E^n , we need at least n vectors for a generator.
- Third, the choice of vectors such as $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ can be made in a large number of alternative ways. In other words, such a set of vectors is not unique. Each of set of such vectors is said to be a **set of generators** for E^2 .

Now, it is clear that we need at least two vectors to generate the vectors in a plane. If these two vectors are linearly independent, we can also assert that any additional vector would be unnecessary for our purpose. In fact, we need exactly two linearly independent vectors to generate all vectors $\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ in E^2 .

Any (two) vectors of E^2 , which are linearly independent and are used to generate the elements of E^2 are **said to constitute the ‘basis’ of E^2** .

In general, we say that a set of vectors $a_1, a_2, \dots, a_n$ from E^n **spans or generates E^n** if

- every vector in E^n can be written as a linear combination of $a_1, a_2, \dots, a_n$.

Further, if we wish to have $a_1, a_2, \dots, a_n$ as a **basis of E^n** , then *additionally*, we should have

- $a_1, a_2, \dots, a_n$ as a linearly independent subset of E^n .

Example: The vector $\mathbf{a} = [2, 3, 4]$ can be written uniquely in terms of the three vectors

$\mathbf{e}_1 = [1, 0, 0]$, $\mathbf{e}_2 = [0, 1, 0]$, and $\mathbf{e}_3 = [0, 0, 1]$ which form a basis for E^3 : $\mathbf{a} = 2\mathbf{e}_1 + 3\mathbf{e}_2 + 4\mathbf{e}_3$

6.8 LET US SUM UP

This unit tells us the way of presenting a variable with magnitude and direction. It also gives the idea about the basic vector operations such as addition, subtraction, and multiplication along with the concepts such as norm, inner product, generator, basis, characteristic equation, eigen value, and eigen vector. The relationship between vector and matrices is covered in this unit.

6.9 KEY WORDS

Basis : Any (two) vectors, which are linearly independent and are used to generate the elements of E^2 are said to constitute the basis of E^2 .

Generator : Any vector $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ can be written as

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = c_1 \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} + c_2 \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} + c_3 \begin{pmatrix} w_1 \\ w_2 \end{pmatrix}, \text{ where}$$

$$\mathbf{u} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}, \mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}, \mathbf{w} = \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} \text{ etc.,}$$

By specifying all possible values for c_1, c_2 and c_3 we may be able to obtain all possible two element vectors $\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$

belonging to the two dimensional space. If this is the case, then we say that $\mathbf{u}, \mathbf{v}$ and $\mathbf{w}$ are the ‘generators’ of the two dimensional spaces (E^2).

Sub Space : w is said to be a subspace of V if

- w is a subset of V and,
- w is a vector space in its own right, i.e., it satisfies all the axioms specified above for a vector space.

Vector Space : Define a set V such that it contains, as its elements, all n – component vectors that can be generated from the field of real numbers, i.e.,
 $V = \{(x_1, x_2, \dots, x_n) \mid x_i \in \mathbf{R}; i = 1, 2, \dots, n\}$. Now if V satisfies the following properties

i) $x, y, z, \in V$, then $x + y \in V$

such that $x + y = y + x$ and,

$$x + (y + z) = (x + Y) + z.$$

V contains a vector 0 such that $x + 0 = x$

ii) For any $\lambda, \theta, \delta \in \mathbf{R}$,

it must be true that $\lambda \cdot x \in V$

Such that $\lambda \cdot (\delta \cdot x) = (\lambda \cdot \delta) \cdot x$, $\lambda \cdot (\delta \cdot x) = \delta (\lambda \cdot x)$
 and $1 \cdot x = x$

iii) $\lambda(\theta + \delta)x = (\lambda \cdot \theta + \lambda \cdot \delta)x$
 and $\lambda(x + y) = \lambda \cdot x + \lambda \cdot y$

Then the set V is said to be a vector space or a linear space over the field $\mathbf{R}$.

Vector : An ordered set expressed in a particular form like (a, b) with a, b as its elements is called a ‘vector’.

6.10 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- $a_1 + 2a_2 + 3a_3 = (5, 3, 7, 9)$
- $2x_2 + 5x_3 - x_1 = (3, -11, -3)$

Check Your Progress 2

- Do yourself
- (i) 9; (ii) 8.
- (i) 43; (ii) -47.

6.11 EXERCISES

Q1. Derive the norm of the following vectors if

(i) $\mathbf{a}=(0 \ 2 \ 1)$; $\mathbf{b}=(6 \ 3 \ 2)$; $\mathbf{c}=(-5 \ 4 \ 3)$; $\mathbf{d}=(0 \ 0 \ 1)$; $\mathbf{e}=(-3 \ -4 \ -2)$.

Ans.: (i) 5; (ii) 49; (iii) 50; (iv) 1; (v) 29.

Q2. If $\mathbf{b} = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, find $\frac{\partial}{\partial \mathbf{x}}(\mathbf{b}'\mathbf{x})$.

Ans.: $\begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$

Q3. If $\mathbf{c} = \begin{bmatrix} c^2 + 1 \\ 3c \\ 4c - 5 \\ c^3 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$, find $\frac{\partial}{\partial \mathbf{x}}(\mathbf{c}'\mathbf{x})$.

Ans.: $\begin{bmatrix} c^2 + 1 \\ 3c \\ 4c - 5 \\ c^3 \end{bmatrix}$

Q4. If $\mathbf{y} = \begin{bmatrix} x_1^2 + 3x_2 \\ 2x_1x_2 - x_2 \\ 2x_1 + x_1x_2 - 3x_2^2 \\ 3x_1^2 - x_1^2x_2 - x_2^3 \end{bmatrix}$, find $\frac{\partial \mathbf{y}}{\partial \mathbf{x}}$.

Ans.: $\begin{bmatrix} 2x_1 & 2x_2 & 2 + x_2 & 6x_1 - 2x_1x_2 \\ 3 & 2x_1 - 1 & x_1 - 6x_2 & -x_1 - 3x_2^2 \end{bmatrix}$

Q5. If $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, $\mathbf{A} = \begin{bmatrix} 3 & 0 & -1 \\ 2 & 1 & 4 \\ -1 & 0 & 3 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$,

find $\frac{\partial}{\partial \mathbf{x}}(\mathbf{x}'\mathbf{A}\mathbf{y})$ and $\frac{\partial}{\partial \mathbf{y}}(\mathbf{x}'\mathbf{A}\mathbf{y})$.

Ans: $\mathbf{y} = \begin{bmatrix} 3y_1 - y_3 \\ 2y_1 + y_2 + 4y_3 \\ -y_1 + 3y_3 \end{bmatrix}$

UNIT 7 VECTORS AND MATRICES

Structure

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7.0 OBJECTIVES

After going through this unit, you should be able to

- present a system of linear equations as a matrix equation;
- explain the concepts of characteristic matrix, characteristic vector; characteristic root, eigen vector, eigen value and explain the characteristic value problem;
- identify the concept of quadratic form;
- present various types of quadratic form;
- discuss the relation of eigen values to definiteness of a quadratic form;
- explain the concept of vector differentiation; and
- describe the processes of various types of vector differentiation.

7.1 INTRODUCTION

In this unit, we shall consider only vectors with a finite number of elements. Further, unless stated otherwise, we shall assume that each element in a vector is a real number. Thus, we are typically concerned with the n -tuples $(x_1, x_2, \dots, x_n)$ of real numbers. A vector all of whose elements are zero is called the '*null vector*'. In economic analysis, quantities of various commodities consumed by an individual can be represented as a vector.

7.2 VECTORS AND MATRICES

A system of ' m ' linear equations in ' n ' variables $(x_1, x_2, \dots, x_n)$ can be arranged into the following format:

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = d_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = d_2$$

.....

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = d_m$$

There are three types of elements in the above equation system. The first is the set of coefficients a_{ij} ; the second is the set of variables $x_1, x_2, \dots, x_n$ and the last is the set of constant terms $d_1, \dots, d_m$. From our earlier discussion in Unit 5 on Matrices, we know that if we arrange the three sets as three rectangular arrays and label them, respectively, as A , X and d (without subscripts), then the above system of equations can be written in the matrix form as

$$AX = d,$$

where,

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}_{m \times n} \quad X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{n \times 1} \quad d = \begin{bmatrix} d \\ d \\ \vdots \\ d_m \end{bmatrix}_{m \times 1}$$

Example: Consider the following linear- equations system

$$6x_1 + 3x_2 + x_3 = 22$$

$$x_1 + 4x_2 - 2x_3 = 12$$

$$4x_1 - x_2 + 5x_3 = 10$$

We can write, this system as the matrix equation

$$\begin{bmatrix} 6 & 3 & 1 \\ 1 & 4 & -2 \\ 4 & -1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 22 \\ 12 \\ 10 \end{bmatrix}.$$

7.3 CHARACTERISTIC VALUE PROBLEM

A problem which arises frequently in application of linear algebra is that of finding values of a scalar parameter λ for which there exists vector $x \neq 0$ (i.e., non-null vector) satisfying $Ax = \lambda x$, where A is a given n th order square matrix. Such a problem is called a **characteristic value ('Eigen Value', or proper value) problem**. If $x \neq 0$ satisfies the equation $Ax = \lambda x$ for a given vector x , then matrix A operating on vector x yields a vector which is a scalar multiple of x .

Thus, λ is called the **characteristic value (or equivalently, characteristic root, Eigen value or proper value)** and x is called the **characteristic vector (or 'Eigen Vector')** associated with A , if $Ax = \lambda x$

7.3.1 Characteristic Equation

For given n , to solve the system of equations $Ax = \lambda x$, for λ , it can be written as $(A - \lambda I)x = 0$, where λI is a unit matrix of order n . Now, if the system has to have a non-trivial solution for the variable vector X , then $\text{rank}[A - \lambda I] < n$, which is true if and only if

$$\begin{vmatrix} a_{11} - \lambda & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} - \lambda & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} - \lambda \end{vmatrix} = 0$$

This gives us an n^{th} degree polynomial equation in λ . This is called the '**characteristic equation**' of A and the left-hand side is called the '**characteristic polynomial**'.

Example: Characteristic polynomial for a second order matrix is

$$\begin{aligned} & \begin{vmatrix} a_{11} - \lambda & a_{12} \\ a_{21} & a_{22} - \lambda \end{vmatrix} = 0 \\ \Rightarrow & (a_{11} - \lambda)(a_{22} - \lambda) - a_{12}a_{21} = 0 \\ \Rightarrow & \lambda^2 - \lambda(a_{11} + a_{22}) + (a_{11}a_{22} - a_{12}a_{21}) = 0 \end{aligned}$$

which is a quadratic equation in λ .

[Note: if A is of order n , then the equation $[A - \lambda I] = 0$ has n roots which may be real or complex, distinct or multiple, and zero or non-zero.]

7.3.2 Sum and Product of Roots

Consider the equation $\lambda^2 - \lambda(a_{11} + a_{22}) + (a_{11}a_{22} - a_{12}a_{21}) = 0$ and let us write it as $c_0 \lambda^2 + c_1 \lambda + c_2 = 0$ where, $c_0 = 1$, $c_1 = -(a_{11} + a_{22})$, $c_2 = a_{11}a_{22} - a_{12}a_{21}$

Applying the knowledge of determining roots of the quadratic equation, we can write,

$$\lambda_1 + \lambda_2 = -\frac{c_1}{c_0} = a_{11} + a_{22}$$

$$\text{and } \lambda_1 \lambda_2 = \frac{c_2}{c_0} = a_{11}a_{22} - a_{21}a_{12} = |A|$$

In general, if A is a square matrix of order n, then the characteristic equation $[A - \lambda I] = 0$ can be written as $c_0 \lambda^n + c_1 \lambda^{n-1} + c_2 \lambda^{n-2} + \dots + c_{n-1} \lambda + c_n = 0$

$$\text{where } c_0 = 1, c_1 = -\sum_{i=1}^n a_{ii}, \text{ and } c_n = (-1)^n |A|$$

C_2 = Sum of all principal minors of A of order 2.

$$C_j = (-1)^j \sum \text{all principal minor of order } j$$

Thus, if characteristic roots are $\lambda_1, \lambda_2, \dots, \lambda_n$,

$$\text{then } \sum_{i=1}^n \lambda_i = \sum_{i=1}^n a_{ii} = \text{Trace}(A) \text{ and } \lambda_1 \cdot \lambda_2 \cdot \dots \cdot \lambda_n = \prod_{i=1}^n \lambda_i = |A|$$

where Trace of a square matrix is the sum of its diagonal elements.

Example: Let, $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$

The characteristic equation is

$$[A - \lambda I] = \begin{vmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{vmatrix} = (2 - \lambda)^2 - 1 = 0$$

$$\text{or, } 4 - 4\lambda + \lambda^2 - 1 = 0$$

$$\text{or, } \lambda^2 - 4\lambda + 3 = 0$$

$$\lambda = \frac{4 \pm \sqrt{16 - 12}}{2} = \frac{4 \pm 2}{2} = 3, 1 \quad \text{i.e. } \lambda_1 = 3, \lambda_2 = 1$$

Properties:

- i) If A is singular square matrix, i.e., $|A| = 0$, then at least one characteristic root of A is zero.
- ii) If A is symmetric all roots are real, i.e., a real symmetric matrix cannot have complex characteristic roots.
- iii) If $\lambda_1, \lambda_2, \dots, \lambda_n$ are the roots of a non-singular matrix A, then the roots of A^{-1} are $1/\lambda_1, 1/\lambda_2, \dots, 1/\lambda_n$
- iv) **If** A and B are two square matrices of the same order such that $AB=BA$ and their roots are $\lambda_1, \lambda_2, \dots, \lambda_n$ and $\mu_1, \mu_2, \dots, \mu_n$, **then** the roots of AB are $\lambda_i \mu_i$; $i=1, 2, \dots, n$.
- v) The eigen values of a diagonal or a triangular matrix coincide with the diagonal elements.
- vi) If $\lambda_1, \lambda_2, \dots, \lambda_n$ are the roots of A and k is an integer, then the roots of A^k are $\lambda_1^k, \lambda_2^k, \dots, \lambda_n^k$.

7.3.3 Characteristic Vector

Let us now consider the determination of the value of $\mathbf{x}$ of the equation $\mathbf{Ax} = \lambda \mathbf{x}$. Note that λ is chosen in such a way that $[\mathbf{A} - \lambda \mathbf{I}]$ is singular, i.e., its rank is less than n . This ensures that $[\mathbf{A} - \lambda \mathbf{I}]\mathbf{x} = \mathbf{0}$ is a homogeneous system of equations having a non-trivial solution. But this solution may not be unique. We may make it unique through some additional restrictions. Further, one set of non-trivial solution $\mathbf{x}^i$ will correspond root of λ_i .

Note: in the above and following discussion the 'i' in $\mathbf{x}^i$, is NOT used in the sense power of $\mathbf{x}$, but as superfix, just like a suffix.

Example: Let us find the characteristics vector for the matrix $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$

As we derived earlier, $\lambda_1 = 3, \lambda_2 = 1$.

$$\therefore \text{For } \lambda_1 = 3, (\mathbf{A} - \lambda_1 \mathbf{I}) = \begin{bmatrix} 2-3 & 1 \\ 1 & 2-3 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$\text{To get } \mathbf{x}^1 = \begin{bmatrix} x_1^1 \\ x_2^1 \end{bmatrix},$$

From $(\mathbf{A} - \lambda_1 \mathbf{I}) \mathbf{x}^1$, we have

$$\left. \begin{array}{l} -x_1^1 + x_2^1 = 0 \\ x_1^1 - x_2^1 = 0 \end{array} \right\} \Rightarrow \mathbf{x}_1^1 = \mathbf{x}_2^1$$

So, the number of solutions (one equation involving two variables) is infinite.

Let us choose one by setting $\mathbf{x}_1^1 = 1$, so that $\mathbf{x}^1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

Next, take $\lambda_2 = 1$, and proceeding in the same way, we get, $x_2^2 = -x_1^2$

$$\text{Again, setting } x_2^2 = 1, \text{ we get } \mathbf{x}^2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

we could normalize $\mathbf{x}^1$ and $\mathbf{x}^2$, i.e., define these such that $\|\mathbf{x}^1\| = \|\mathbf{x}^2\|$.

$$\text{This would give us } \mathbf{x}^1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \text{ and } \mathbf{x}^2 = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$$

Note: $\mathbf{x}^1$ and $\mathbf{x}^2$ are orthogonal to each other.

7.3.4 Diagonalisation

It is a general property of real symmetric matrices that their roots are always real and that the associated characteristic vectors are mutually orthogonal. This implies that such a matrix can always be diagonalized by $\mathbf{P}$, the matrix of n characteristic vectors.